Iron Acres (BB Edition) Manual

This manual is based on the original Iron Acres manual, with modifications and clarifications for the workshops and competitions that will be held at Bloomberg.

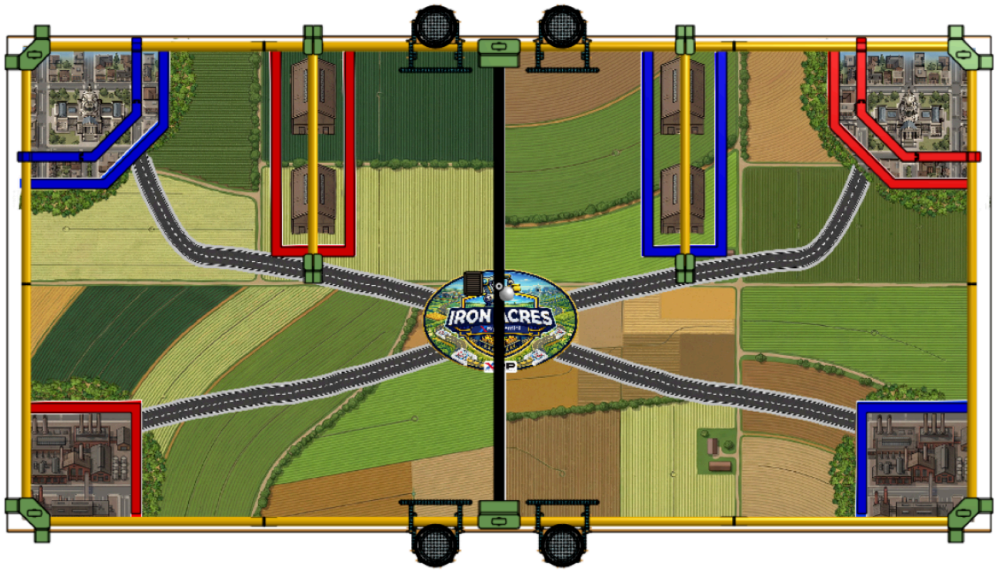
Last Updated: 10/14/2026

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1 Game Overview

In Iron Acres, 2 ALLIANCES (an ALLIANCE is a cooperative of up to 2 teams) are tasked with harvesting CROPS, processing them into PRODUCTS at their FACTORY, and delivering those PRODUCTS to their CITY. Teams score points by tipping SILOS, scoring CROPS in their FACTORY, delivering PRODUCTS to their CITY, and parking in or hanging from their BARN as the match concludes.



1. The challenge will begin with a 10-second AUTONOMOUS period (See STARTING POSITIONS in Section 5)
    1.1. Each SILO tipped during AUTONOMOUS scores 5 points.
    1.2. Each CROP scored in the ALLIANCE's FACTORY during AUTONOMOUS scores 7 points.
    1.3. Robots that are PARKED (even partially) in their ALLIANCE's FACTORY at the end of AUTONOMOUS score an additional 5 points.

2. After the 10-second AUTONOMOUS period, there will be a 10-second transition period for officials to count AUTONOMOUS game pieces and for teams to switch to the TELEOPERATED period. Robots may NOT move and are NOT repositioned during the transition period; they begin TELEOP wherever they were at the end of AUTONOMOUS.

3. The 2-minute 30-second TELEOPERATED period will then begin with the prompt "Drivers, pick up your controllers" and a 3-2-1 countdown.
    3.1. During TELEOP, robots gather CROPS from the field and deliver them to their ALLIANCE's FACTORY. Each CROP scored in the FACTORY is worth 7 points.
    3.2. For every 4 CROPS scored in the FACTORY, the HUMAN PLAYER may move those CROPS into FACTORY STORAGE and exchange them for 1 PRODUCT, which they place on a robot in the FACTORY.

3.3. Robots then deliver PRODUCTS to their ALLIANCE's CITY, located on the opposite half of the field.

3.3.1. Each PRODUCT in the CITY LIMITS at the end of the match scores 10 points.

3.3.2. Each PRODUCT in the CITY CENTER at the end of the match scores 15 points.

4. The last 30 seconds of TELEOP is the ENDGAME.

4.1. Robots that are PARKED (even partially) in their ALLIANCE's BARN at the end of the match score 5 points.

4.2. Robots that are HANGING from their ALLIANCE's BARN BAR at the end of the match score 25 points. A HANGING robot does not also earn BARN PARKING points.

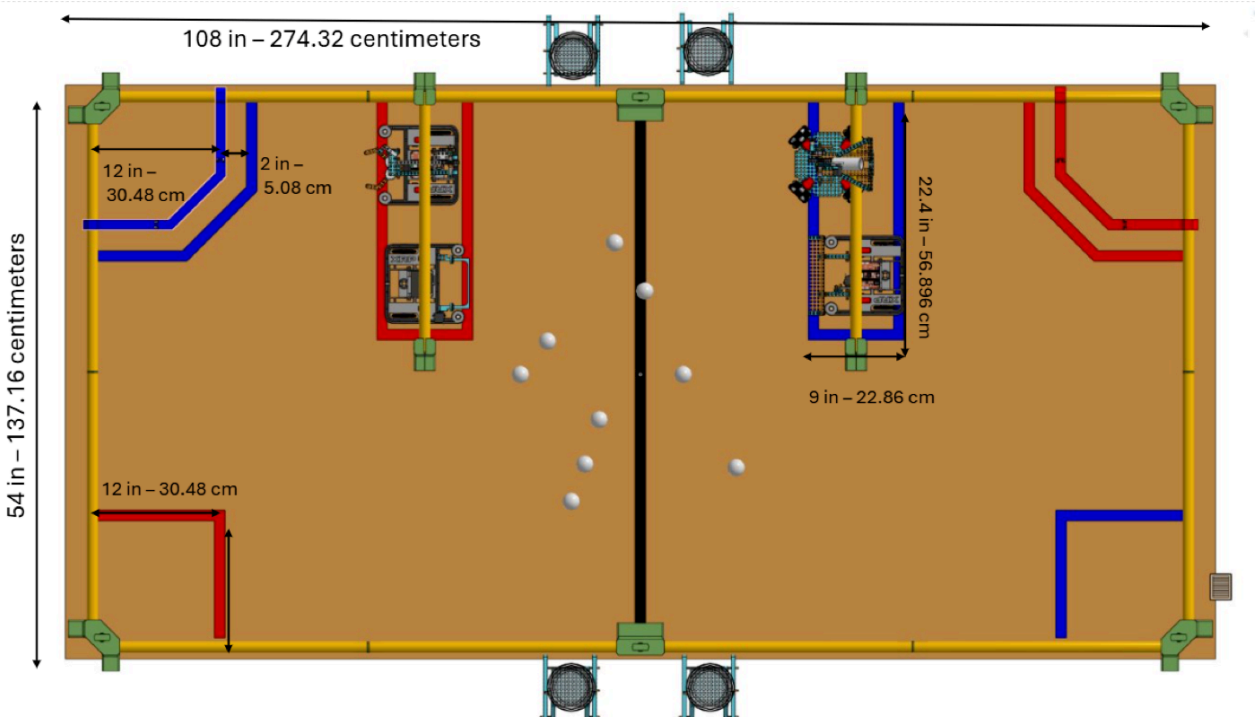
Scoring Summary

		MATCH Points	
		AUTO	**TELEOP**
SILO	SILO tipped	5	
GAME PIECE	CROP scored in the FACTORY	7	7
	PRODUCT in the CITY LIMITS		10
	PRODUCT in the CITY CENTER		15
PARK	PARKED in the FACTORY of your ALLIANCE color	5	
	PARKED in the BARN of your ALLIANCE color		5
HANG	HANGING from the BARN BAR of your ALLIANCE color		25

Scoring Notes

1. AUTONOMOUS points (SILOS, FACTORY CROPS, FACTORY PARKING) are assessed during the transition period at the end of AUTONOMOUS.
2. SILOS only score when tipped during AUTONOMOUS. SILOS may still be tipped during TELEOP to release CROPS, but score no points.
3. PRODUCT, BARN PARKING, and HANGING points are assessed once the match has ended and all robots and game pieces have come to rest.
4. Each CROP is counted only once.
    - 4.1. CROPS scored in AUTONOMOUS are not counted again in TELEOP.
    - 4.2. CROPS that the HUMAN PLAYER moves from the FACTORY into FACTORY STORAGE (including CROPS exchanged for PRODUCTS) remain scored.
5. A robot is PARKED in a location if any part of its chassis is within the tape boundary of that location.
6. A robot is HANGING if it is fully supported by the BARN BAR and is not touching the field floor.
7. PRODUCT scoring
    - 7.1. A PRODUCT inside the CITY WALL scores as CITY CENTER.
    - 7.2. A PRODUCT in the CITY LIMITS scores as CITY LIMITS.
    - 7.3. A PRODUCT resting on the CITY WALL scores as CITY CENTER if it is not in contact with the field floor, and as CITY LIMITS if it is in contact with the field floor.

2 Field Overview



The Iron Acres field is 108" x 54" and is divided into a RED half and a BLUE half by the CENTERLINE. Each ALLIANCE's FACTORY and BARN are on its own half of the field. Each ALLIANCE's CITY is on the opposing ALLIANCE's half of the field.

Scoring Locations

The Iron Acres field contains several scoring locations.

1. FACTORY
    - 1.1. Each ALLIANCE has a FACTORY: a 12" x 12" square in a corner of its own half of the field, outlined in ALLIANCE-colored tape.
    - 1.2. FACTORY STORAGE is a container, located outside the field next to the FACTORY, used by the HUMAN PLAYER to hold CROPS that have been processed. A spare SILO tub may be used.

2. CITY
    - 2.1. Each ALLIANCE has a CITY in a corner of the opposing ALLIANCE's half of the field, outlined in ALLIANCE-colored tape.
    - 2.2. The CITY WALL is a 2" tall boundary wall that measures 12" x 12" to its outside edge.
    - 2.3. The CITY CENTER is the area inside the CITY WALL.
    - 2.4. The CITY PERIMETER is a tape line on the floor, 2" outside of the CITY WALL.
    - 2.5. The CITY LIMITS is the 2" area between the CITY WALL and the CITY PERIMETER.

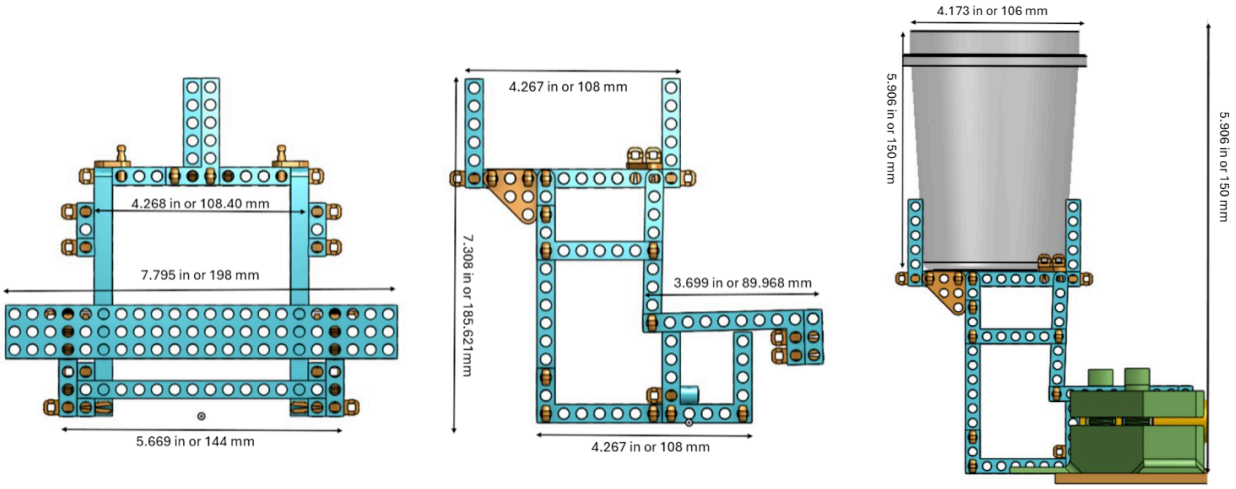
3. SILOS

3.1. There are 4 SILOS. Each ALLIANCE has 2 SILOS on its own half of the field, mounted outside the field wall near the CENTERLINE.

3.2. Each SILO is a tub that sits on top of a SILO HOLDER. At the start of the match, each SILO holds 10 CROPS, minus any CROPS that robots of that ALLIANCE have preloaded.

3.3. The SILO HOLDER is built from XBS components and is 7.3" tall x 7.795" wide. Pushing its front plate tips the SILO, dumping its CROPS onto the field.

3.4. The SILO tub is 4.173" in diameter and 5.906" tall (a deli-style storage container; the exact size may vary slightly).

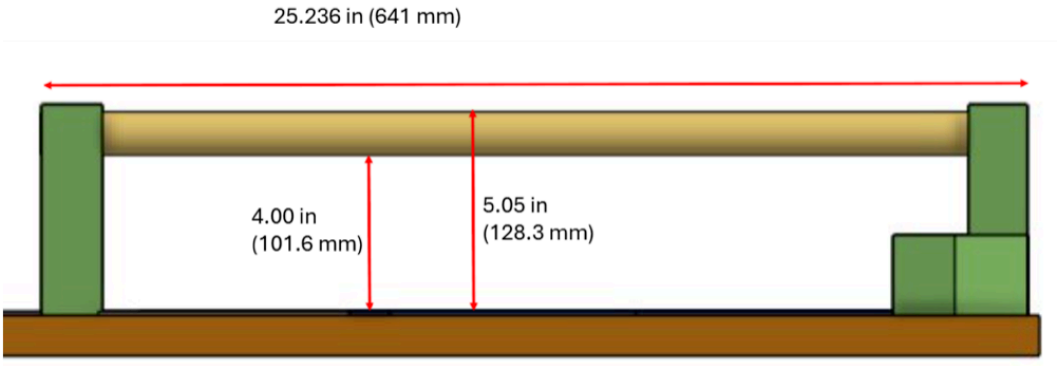


SILO HOLDER SILO on SILO HOLDER

4. BARN

4.1. Each ALLIANCE has a BARN on its own half of the field: a 9" x 22.4" area outlined in ALLIANCE-colored tape. Robots start the match in their BARN.

4.2. The BARN BAR spans the BARN. It is 25.236" wide; the bottom of the bar is 4.00" above the field floor and the top of the bar is 5.05" above the field floor.



BARN BAR

Other Field Locations

1. CENTERLINE
    1.1. The CENTERLINE is a 1" wide line that divides the field into a RED half and a BLUE half (See image).

Game Pieces

1. CROP
    1.1. Each CROP is a white ball with a diameter of 40-43 mm (a table tennis ball or a wiffle golf ball).
    1.2. There are 40 CROPS in total. At the start of the match, every CROP is either in a SILO or preloaded on a robot.

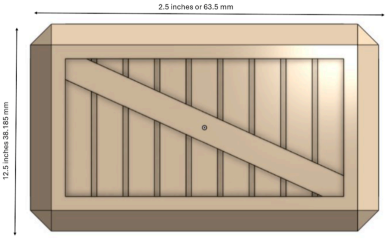
2. PRODUCT
    2.1. Each PRODUCT is a rectangular 3D print measuring 1.5" x 2.5".
    2.2. PRODUCTS are held by the HUMAN PLAYER and enter the field only when the HUMAN PLAYER exchanges CROPS for them.



CROPS (table tennis balls)



CROP (wiffle golf ball)



PRODUCT (1.5" x 2.5")

3 Robot Construction Rules

1. Each robot must follow these rules in order to pass inspection and compete.
    1.1. A robot must fit inside a 12" x 12" x 12" cube **at all times** during the match (not just in its starting configuration).

2. A robot may not contain materials or components that could cause damage to the playing field.

3. A robot should have a method of safely storing energy; for example, no launching rubber bands.

4. Robots must prominently display their ALLIANCE color and team number during matches
    4.1. Acceptable markers
        4.1.1. Blue or Red tape with team number clearly visible
        4.1.2. Clip on indicators (either team provided, or provided by event organizers)

5. Note: Additional rules may be necessary as the workshop season progresses. These will be published in a separate update document.

4 Penalties

Breaching the following rules will result in a PENALTY or a RED CARD

1. PENALTY (10 points awarded to the opposing ALLIANCE)
    - 1.1. PINNING a robot for 3 or more seconds (1 additional PENALTY for every further 3 seconds of PINNING)
        - 1.1.1. PINNING is holding another robot against a field element so that it cannot escape.
    - 1.2. During AUTONOMOUS:
        - 1.2.1. Drive teams may not interact with the robot or operator controls. An ALLIANCE that violates this rule also receives 0 AUTONOMOUS points.
        - 1.2.2. Robots may not cross the CENTERLINE.
    - 1.3. Robots may not contact or block an opposing robot that is in its own FACTORY or CITY (1 PENALTY per occurrence)
    - 1.4. Robots may not remove CROPS or PRODUCTS from the opposing ALLIANCE's FACTORY or CITY (1 PENALTY per game piece removed)
    - 1.5. Robots may possess no more than 2 PRODUCTS at a time (1 PENALTY per occurrence). There is no limit to the number of CROPS a robot may possess.
    - 1.6. HUMAN PLAYERS:
        - 1.6.1. May only interact with game pieces in their own ALLIANCE's FACTORY and FACTORY STORAGE
        - 1.6.2. May place PRODUCTS on robots, but may not otherwise manipulate CROPS or PRODUCTS on a robot
        - 1.6.3. May only introduce 1 PRODUCT for every 4 CROPS moved into FACTORY STORAGE. For example, if an ALLIANCE has 3 PRODUCTS on the field, it must have at least 12 CROPS in FACTORY STORAGE.
    - 1.7. A robot or HUMAN PLAYER may not launch, shoot or throw a game piece
    - 1.8. A robot may not damage or functionally impair another robot or the playing field
    - 1.9. Robots may not intentionally detach or leave parts on the field (1 PENALTY per match)
    - 1.10. No one can enter the field during a match
2. RED CARD (Disqualification from the match)
    - 2.1. Excessive, ungracious, or unprofessional behavior will result in a red card
    - 2.2. Throwing CROPS or PRODUCTS at anyone is considered an automatic red card
    - 2.3. The team must send at least 1 team member to a match, whether the robot is there or not
    - 2.4. Repeated, egregious, or strategic violations of PENALTY-able offenses (for example, intentionally damaging robots or repeatedly crossing the CENTERLINE in AUTONOMOUS) may result in a RED CARD

5 Match Play

Match Setup

1. During match setup, teams put their robots in their STARTING POSITIONS and may preload up to 2 CROPS per robot.
    1.1. Preloaded CROPS are taken from that ALLIANCE's SILOS, so the SILOS will hold fewer CROPS.
    1.2. Please make sure to avoid stepping on the field elements while putting robots onto the field.
2. STARTING POSITIONS
    2.1. Robots begin the match fitting inside a 12" x 12" x 12" cube
    2.2. Robots must start with their wheels inside the BARN of their ALLIANCE color. The rest of the robot may overhang the BARN tape.
3. Each ALLIANCE may have 1 HUMAN PLAYER, positioned outside the field next to their FACTORY.

AUTONOMOUS Period

1. Each match begins with a 10-second AUTONOMOUS period.
    1.1. A sound will play to signal that the AUTONOMOUS period has started.
    1.2. At the start of the AUTONOMOUS period, each driver should switch to the Autonomous mode in the Simulator Driver Station.
    1.3. After that, drivers may not touch their controls until the start of the TELEOP period.
    1.4. A sound will play to signal the END of AUTONOMOUS. At this point, drivers should switch to the Disabled mode in the Simulator Driver Station.

Transition from AUTONOMOUS to TELEOP

1. After AUTONOMOUS is completed, there will be a 10-second transition period for referees and scorers to complete AUTONOMOUS scoring, and for drivers to prepare for TELEOP. Robots may NOT move, and robots and game pieces are NOT repositioned.

TELEOP Period

1. After the transition period, a 2-minute 30-second TELEOP period will begin.
    1.1. The announcer will say "Drivers, pick up your controllers" and count down 3-2-1, and a sound will play to signal that the TELEOP period has begun.
    1.2. During TELEOP, drivers are free to touch their controls.
    1.3. Once there are 30 seconds left in the TELEOP period, a sound will play to signal that the ENDGAME phase has begun.
    1.4. A sound will play to signal that TELEOP has ended. Drivers must then stop providing input to their robots, and switch to Disabled in the Simulator Driver Station.

6 Competition Setup

The competition will consist of randomly selected 2 vs 2 matches. The number of matches to be played will vary depending on how many teams are registered. The team with the highest average score after the final match is declared the event winner.

Judges will also be interviewing teams to ask about their approach to building a robot for the challenge. Awards may also be given out to teams based on programming, mechanical design, and other factors as a result of these interviews.

Change Log

Date	Change Description
10/14/2026	Initial Publication